

tkinter库GUI编程

tkinter库（内置库，稳定）适合初学者进行简单GUI编程，但需手动，且控件相对较少。
大型、成熟功能复杂的系统可以用PySide/PyQt (拖拽式控件，自动变成py代码)

库名	上手难度	界面美观度	适用场景
Tkinter	低	一般	学习、小工具
PyQt6	中等	高	商业软件、复杂应用
Dear PyGui	中等	高	数据可视化、调试工具
Flet	低	高	跨平台现代应用
Kivy	中等	高	移动端、触摸屏

Tk, Frame

特性	Tk()	Frame()
是什么	主窗口 / 根窗口	容器 / 面板
数量	只能 1 个	可以无数个
能否独立存在	能，独立窗口	不能，必须放在 Tk 或其他 Frame 里
有无标题栏	有	无（默认）
核心用途	程序主界面	分区、排版、分组

```
import tkinter as tk

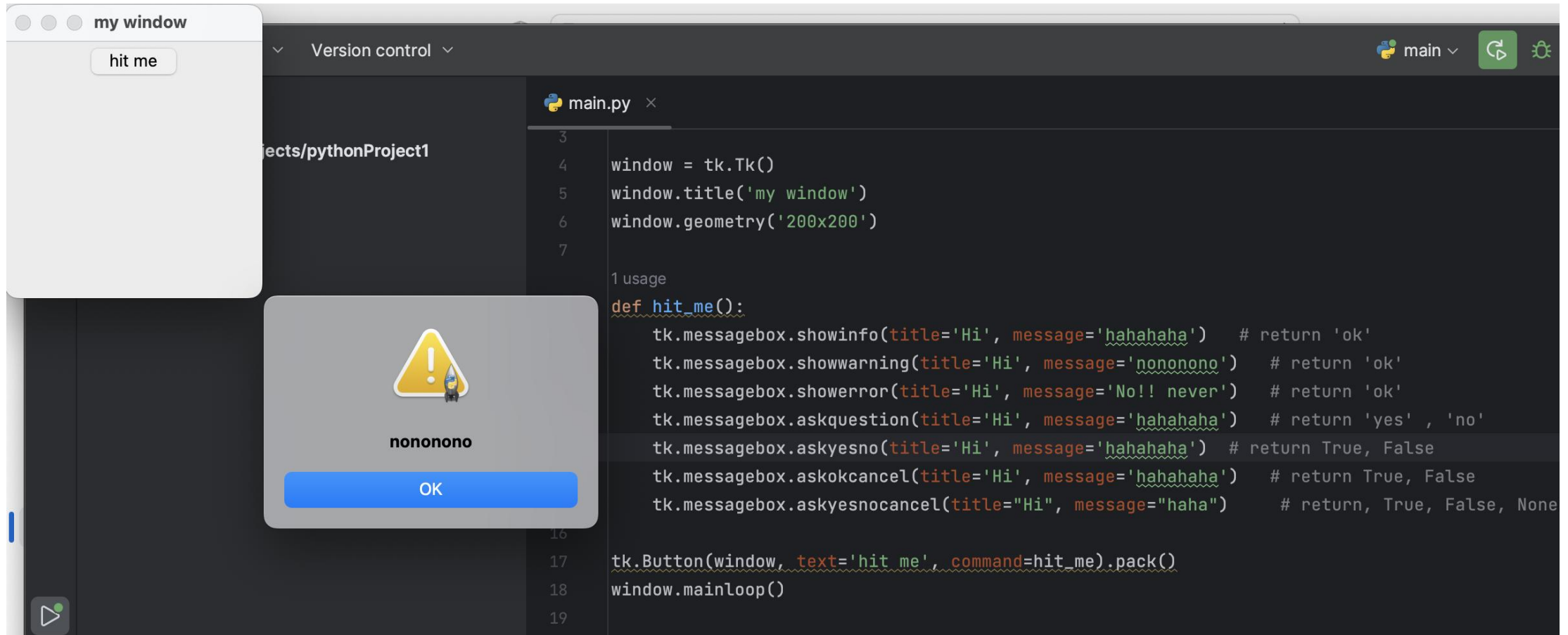
window = tk.Tk()
window.title('my window')
window.geometry('200x200')
tk.Label(window, text='on the window').pack()

frm = tk.Frame(window)
frm.pack()
frm_l = tk.Frame(frm, )
frm_r = tk.Frame(frm)
frm_l.pack(side='left')
frm_r.pack(side='right')

tk.Label(frm_l, text='on the frm_l1').pack()
tk.Label(frm_l, text='on the frm_l2').pack()
tk.Label(frm_r, text='on the frm_r1').pack()
window.mainloop()
```



messagebox



pack, place, grid方法放置控件

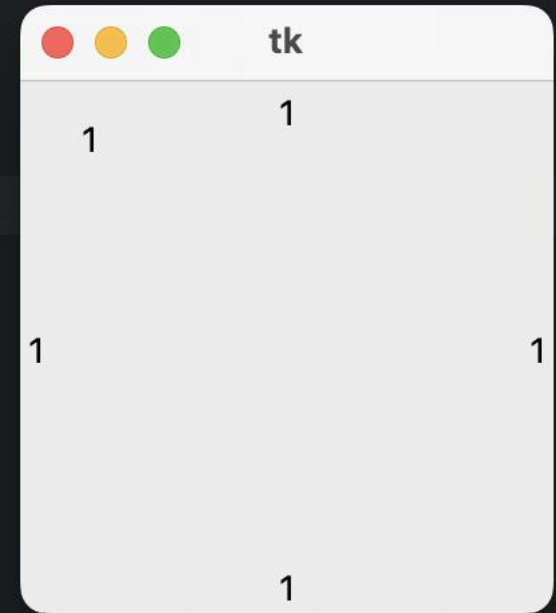
```
import tkinter as tk

window = tk.Tk()
window.geometry('200x200')

tk.Label(window, text='1').pack(side='top')
tk.Label(window, text='1').pack(side='bottom')
tk.Label(window, text='1').pack(side='left')
tk.Label(window, text='1').pack(side='right')

tk.Label(window, text=1).place(x=20, y=10, anchor='nw')

window.mainloop()
```



#四行三列放置用 grid

for i in range(4):

for j in range(3):

tk.Label(window, text=1).grid(row=i, column=j, padx=10, pady=10)

Label, Button

```
import tkinter as tk
window = tk.Tk()
window.title('my window')
window.geometry('200x100')
var = tk.StringVar()
l = tk.Label(window, textvariable=var, bg='green', font=('Arial', 12), width=15, height=2)
l.pack()

on_hit = False

1 usage
def hit_me():
    global on_hit
    if on_hit == False:
        on_hit = True
        var.set('you hit me')
    else:
        on_hit = False
        var.set('')

b = tk.Button(window, text='hit me', width=15, height=2, command=hit_me)
b.pack()

window.mainloop()
```



Entry

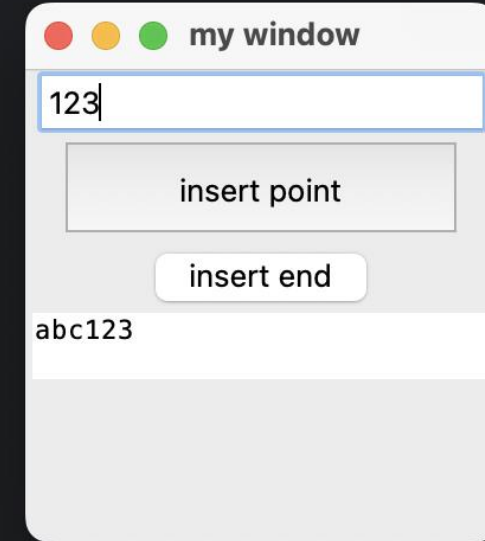
```
import tkinter as tk

window = tk.Tk()
window.title('my window')
window.geometry('200x200')
# e = tk.Entry(window, show="*")
e = tk.Entry(window)
e.pack()

1 usage
def insert_point():
    var = e.get()
    t.insert('insert', var)

1 usage
def insert_end():
    var = e.get()
    # t.insert('end', var)
    t.insert(2.2, var)

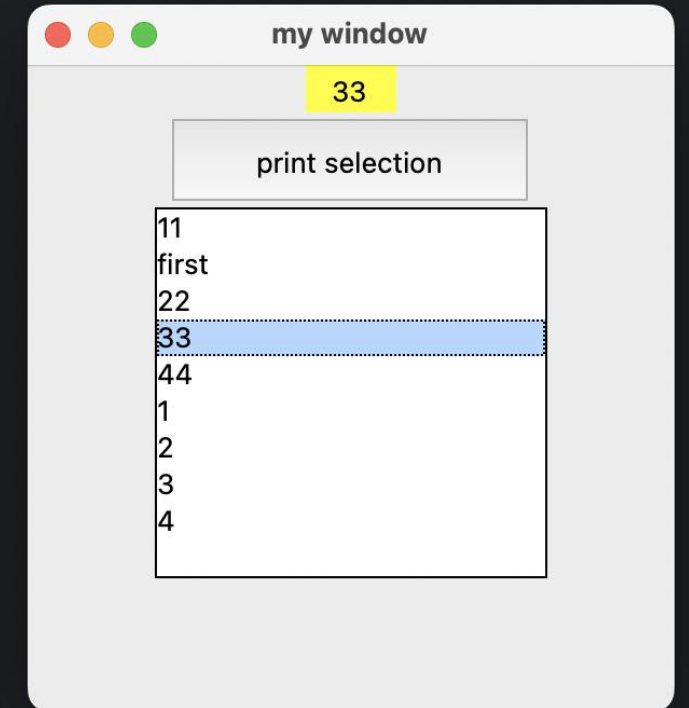
b1 = tk.Button(window, text='insert point', width=15,
               height=2, command=insert_point)
b1.pack()
b2 = tk.Button(window, text='insert end',
               command=insert_end)
b2.pack()
t = tk.Text(window, height=2)
t.pack()
window.mainloop()
```



Listbox

```
import tkinter as tk
window = tk.Tk()
window.title('my window')
window.geometry('300x300')
var1 = tk.StringVar()
l = tk.Label(window, bg='yellow', width=4, textvariable=var1)
l.pack()
1 usage
def print_selection():
    value = lb.get(lb.curselection())
    var1.set(value)

b1 = tk.Button(window, text='print selection', width=15,
                height=2, command=print_selection)
b1.pack()
var2 = tk.StringVar()
var2.set((11,22,33,44))
lb = tk.Listbox(window, listvariable=var2)
list_items = [1,2,3,4]
for item in list_items:
    lb.insert('end', item)
lb.insert(1, 'first')
lb.insert(2, 'second')
lb.delete(2)
lb.pack()
window.mainloop()
```



Radiobutton

```
import tkinter as tk
window = tk.Tk()
window.title('my window')
window.geometry('200x200')
var = tk.StringVar()
l = tk.Label(window, bg='yellow', width=20, text='empty')
l.pack()

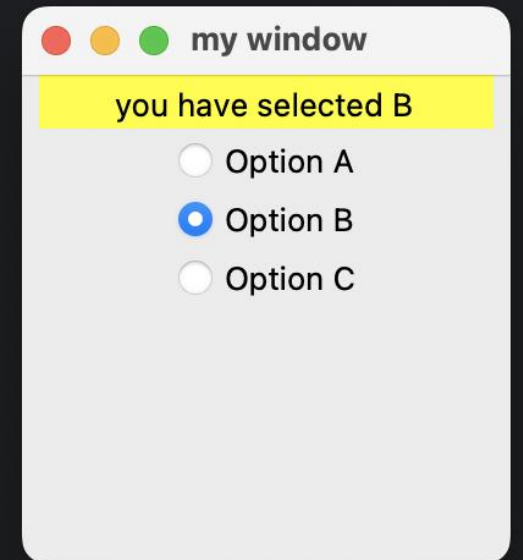
3 usages
def print_selection():
    l.config(text='you have selected ' + var.get())

r1 = tk.Radiobutton(window, text='Option A',
                    variable=var, value='A',
                    command=print_selection)
r1.pack()

r2 = tk.Radiobutton(window, text='Option B',
                    variable=var, value='B',
                    command=print_selection)
r2.pack()

r3 = tk.Radiobutton(window, text='Option C',
                    variable=var, value='C',
                    command=print_selection)
r3.pack()

window.mainloop()
```



Scale

```
import tkinter as tk

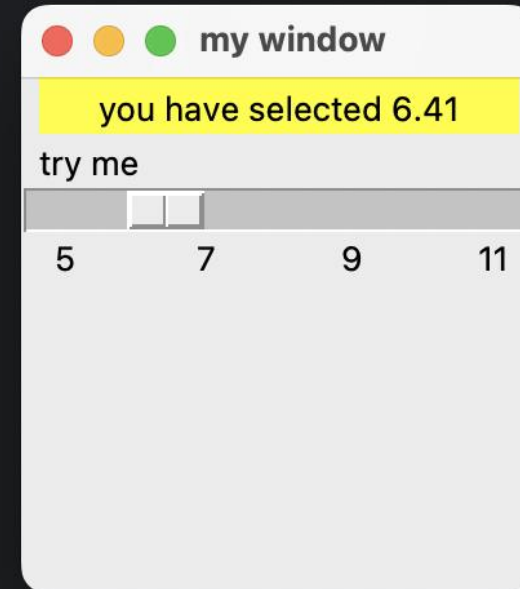
window = tk.Tk()
window.title('my window')
window.geometry('200x200')

l = tk.Label(window, bg='yellow', width=20, text='empty')
l.pack()

1 usage
def print_selection(v):
    l.config(text='you have selected ' + v)

s = tk.Scale(window, label='try me', from_=5, to=11, orient=tk.HORIZONTAL,
              length=200, showvalue=0, tickinterval=2, resolution=0.01, command=print_selection)
s.pack()

window.mainloop()
```



Checkbutton

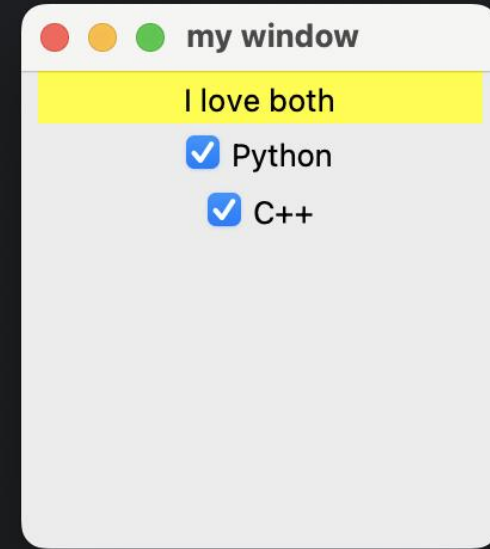
```
import tkinter as tk

window = tk.Tk()
window.title('my window')
window.geometry('200x200')
l = tk.Label(window, bg='yellow', width=20, text='empty')
l.pack()

2 usages
def print_selection():
    if (var1.get() == 1) & (var2.get() == 0):
        l.config(text='I love only Python ')
    elif (var1.get() == 0) & (var2.get() == 1):
        l.config(text='I love only C++')
    elif (var1.get() == 0) & (var2.get() == 0):
        l.config(text='I do not love either')
    else:
        l.config(text='I love both')

var1 = tk.IntVar()
var2 = tk.IntVar()
c1 = tk.Checkbutton(window, text='Python', variable=var1, onvalue=1, offvalue=0,
                    command=print_selection)
c2 = tk.Checkbutton(window, text='C++', variable=var2, onvalue=1, offvalue=0,
                    command=print_selection)

c1.pack()
c2.pack()
window.mainloop()
```

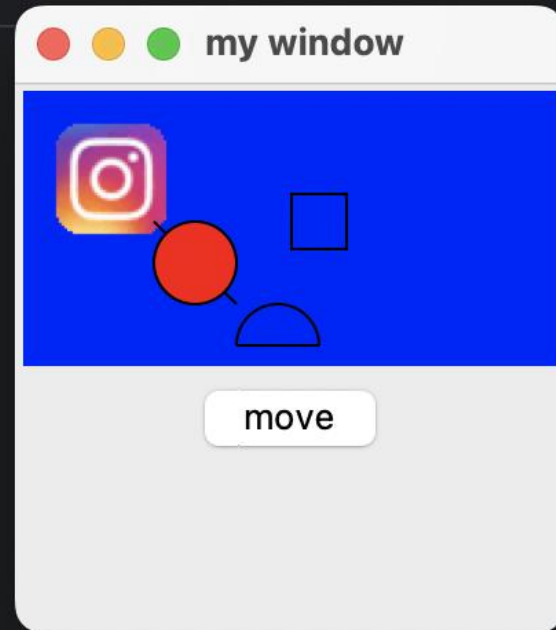


Canvas

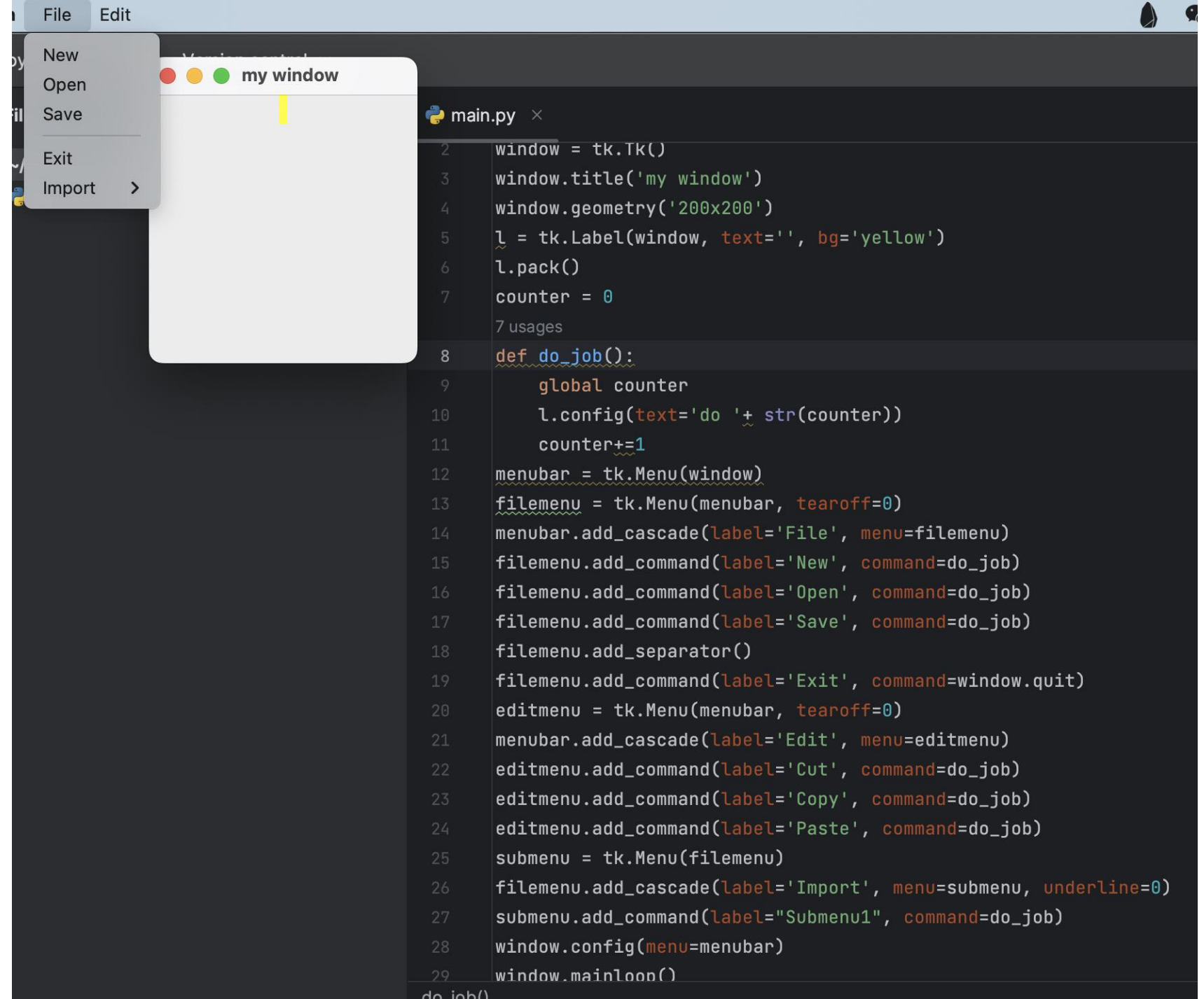
```
import tkinter as tk
window = tk.Tk()
window.title('my window')
window.geometry('200x200')
canvas = tk.Canvas(window, bg='blue', height=100, width=200)
image_file = tk.PhotoImage(file='ins.gif')
image = canvas.create_image(10, 10, anchor='nw', image=image_file)
x0, y0, x1, y1 = 50, 50, 80, 80
line = canvas.create_line(x0, y0, x1, y1)
oval = canvas.create_oval(x0, y0, x1, y1, fill='red')
arc = canvas.create_arc(x0+30, y0+30, x1+30, y1+30, start=0, extent=180)
rect = canvas.create_rectangle(100, 30, 100+20, 30+20)
canvas.pack()

1 usage
def moveit():
    canvas.move(rect, 0, 2)

b = tk.Button(window, text='move', command=moveit).pack()
window.mainloop()
```



Menu



登陆界面

```
as tk
port messagebox # import this to fix

)
elcome to Mofan Python')
('450x300')

vas(window, height=200, width=500)
.PhotoImage(file='welcome.gif')
create
e='top

ion
, text
, text

tk.Str
t('exa
= tk.E
place(x=100, y=100)
k.StringVar()
```

Welcome to Mofan Python



User name:

Password:

Sign up window

User name:

Password:

Confirm password:


```

import tkinter as tk

from tkinter import messagebox # import this to fix messagebox error
import pickle

window = tk.Tk()
window.title('Welcome to Mofan Python')
window.geometry('450x300')

# welcome image
canvas = tk.Canvas(window, height=200, width=500)
image_file = tk.PhotoImage(file='welcome.gif')
image = canvas.create_image(0,0, anchor='nw', image=image_file)
canvas.pack(side='top')

# user information
tk.Label(window, text='User name: ').place(x=50, y=150)
tk.Label(window, text='Password: ').place(x=50, y=190)

var_usr_name = tk.StringVar()
var_usr_name.set('example@python.com')
entry_usr_name = tk.Entry(window, textvariable=var_usr_name)
entry_usr_name.place(x=160, y=150)
var_usr_pwd = tk.StringVar()
entry_usr_pwd = tk.Entry(window, textvariable=var_usr_pwd, show='*')
entry_usr_pwd.place(x=160, y=190)

```

```

def usr_login():
    usr_name = var_usr_name.get()
    usr_pwd = var_usr_pwd.get()
    try:
        with open('usr_info.pickle', 'rb') as usr_file:
            usr_info = pickle.load(usr_file)
    except FileNotFoundError:
        with open('usr_info.pickle', 'wb') as usr_file:
            usr_info = {'admin': 'admin'}
            pickle.dump(usr_info, usr_file)
    if usr_name in usr_info:
        if usr_pwd == usr_info[usr_name]:
            tk.messagebox.showinfo(title='Welcome', message='How are you? ' + usr_name)
        else:
            tk.messagebox.showerror(message='Error, your password is wrong, try again.')
    else:
        is_sign_up = tk.messagebox.askyesno('Welcome',
                                             'You have not signed up yet. Sign up today?')
        if is_sign_up:
            usr_sign_up()

```

```

def usr_sign_up():
    np = new_pwd.get()
    npf = new_pwd_confirm.get()
    nn = new_name.get()
    with open('usr_info.pickle', 'rb') as usr_file:
        exist_usr_info = pickle.load(usr_file)
    if np != npf:
        tk.messagebox.showerror('Error', 'Password and confirm password must be the same!')
    elif nn in exist_usr_info:
        tk.messagebox.showerror('Error', 'The user has already signed up!')
    else:
        exist_usr_info[nn] = np
        with open('usr_info.pickle', 'wb') as usr_file:
            pickle.dump(exist_usr_info, usr_file)
        tk.messagebox.showinfo('Welcome', 'You have successfully signed up!')
        window_sign_up.destroy()
window_sign_up = tk.Toplevel(window)
window_sign_up.geometry('350x200')
window_sign_up.title('Sign up window')

```

```

new_name = tk.StringVar()
new_name.set('example@python.com')
tk.Label(window_sign_up, text='User name: ').place(x=10, y=10)
entry_new_name = tk.Entry(window_sign_up, textvariable=new_name)
entry_new_name.place(x=150, y=10)

```

```

new_pwd_confirm = tk.StringVar()
tk.Label(window_sign_up, text='Confirm password: ').place(x=10, y=90)
entry_usr_pwd_confirm = tk.Entry(window_sign_up, textvariable=new_pwd_confirm, show='*')
entry_usr_pwd_confirm.place(x=150, y=90)

```

```

btn_confirm_sign_up = tk.Button(window_sign_up, text='Sign up', command=sign_to_Mofan_Python)
btn_confirm_sign_up.place(x=150, y=130)

```

```

# login and sign up button
btn_login = tk.Button(window, text='Login', command=usr_login)
btn_login.place(x=170, y=230)
btn_sign_up = tk.Button(window, text='Sign up', command=usr_sign_up)
btn_sign_up.place(x=270, y=230)

window.mainloop()

```